EVIDENCE-FIRST PORTFOLIO

Quantum / EXON Red-Team Case Study

Speculative physics handled as bounded negative-result research.

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What it shows

The Quantum/EXON material is included as disciplined red-team evidence, not as a breakthrough claim. I pushed a speculative idea through no-signaling criteria and preserved the negative result.

Under standard linear quantum mechanics and local CPTP maps, Bob-local distinguishability does not appear. Any positive toy model requires nonstandard physics and is not a validated channel.

Reasoning sequence

Step	Question	Outcome
Hypothesis	Can Alice's choice create a locally readable Bob-side difference?	Only meaningful if Bob can distinguish local states without postselection or side information.
Standard-QM model	Do local operations create a trace-distance difference on Bob's reduced state?	No. Local CPTP dynamics preserve no-signaling.
Intermediate-mode idea	Can a resonator or pointer make the difference visible?	Not under standard assumptions.
New-physics cracks	Could nonlinear or hidden-variable alternatives create distinguishability?	Only by leaving standard quantum mechanics and accepting severe constraints.
Portfolio conclusion	What is the honest claim?	A rigorous negative-result boundary study.

What I learned

The value of this project is not that it succeeded. The value is that I let it fail correctly. I moved from an ambitious hypothesis into formal criteria, then accepted the boundary when the standard theorem closed the path.